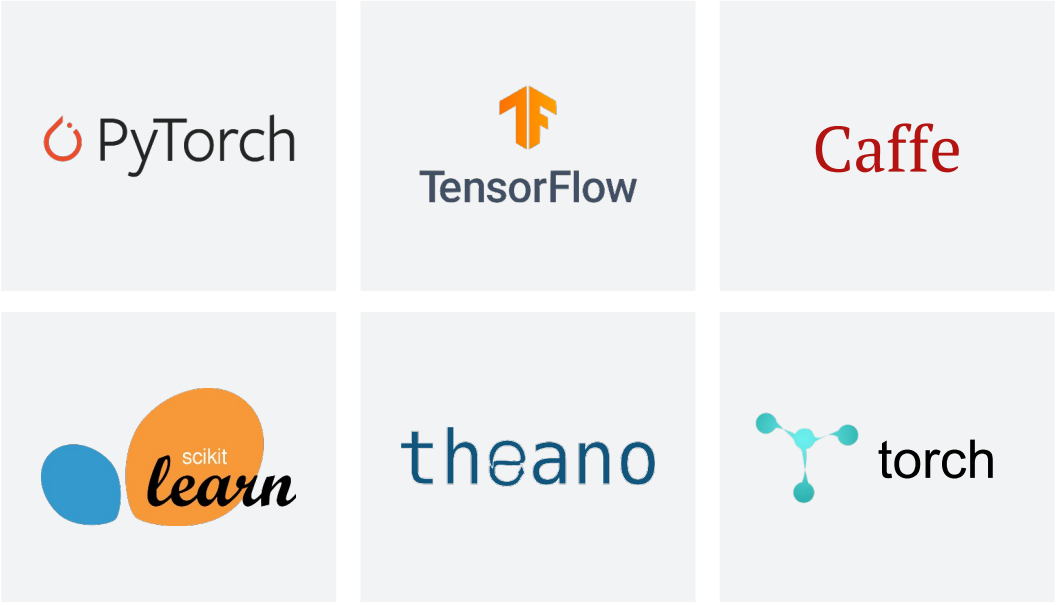
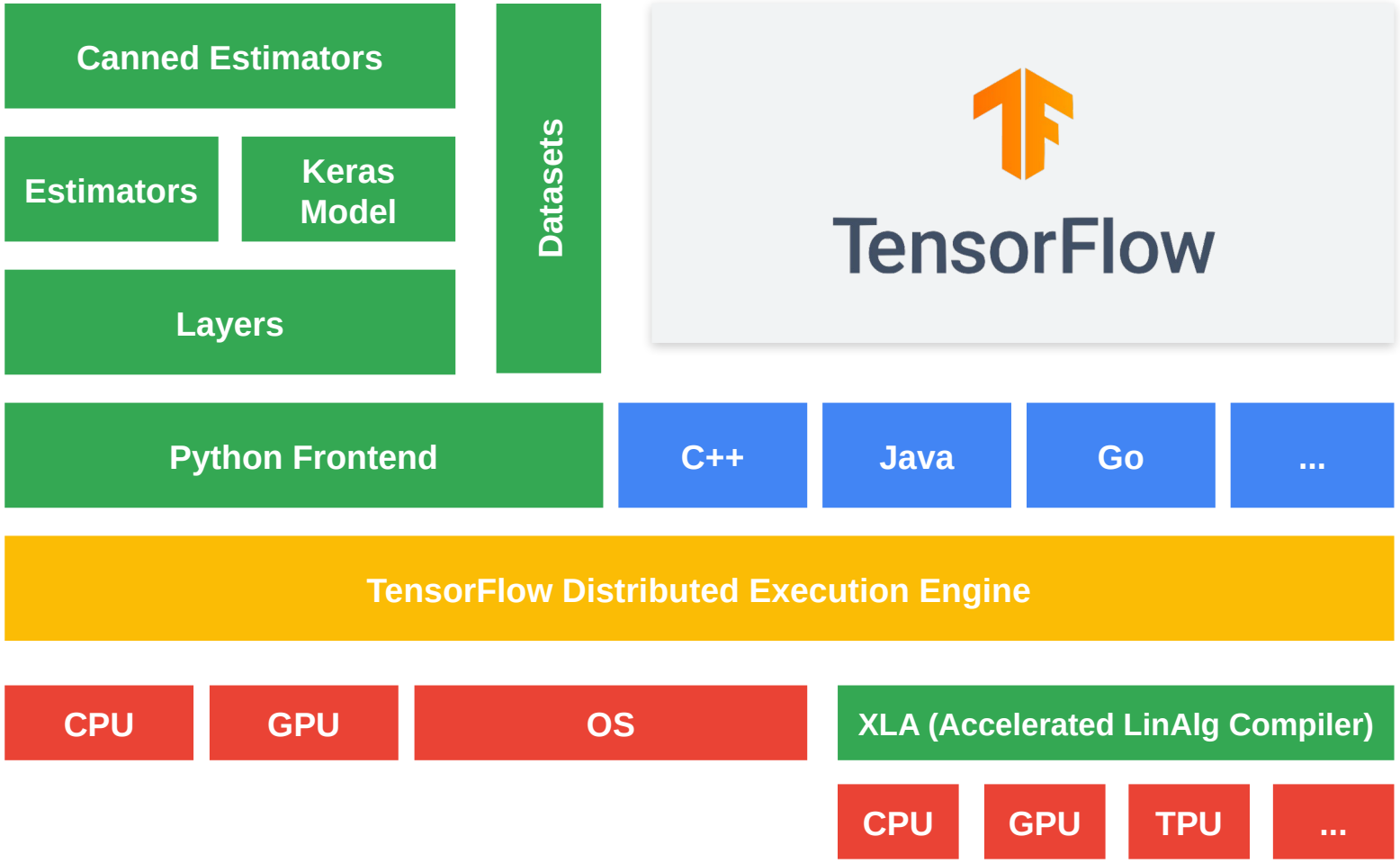
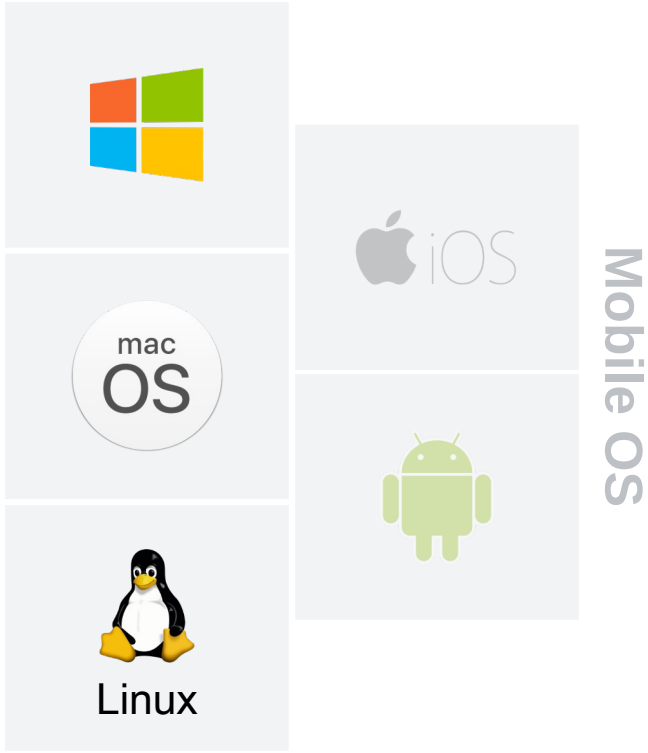
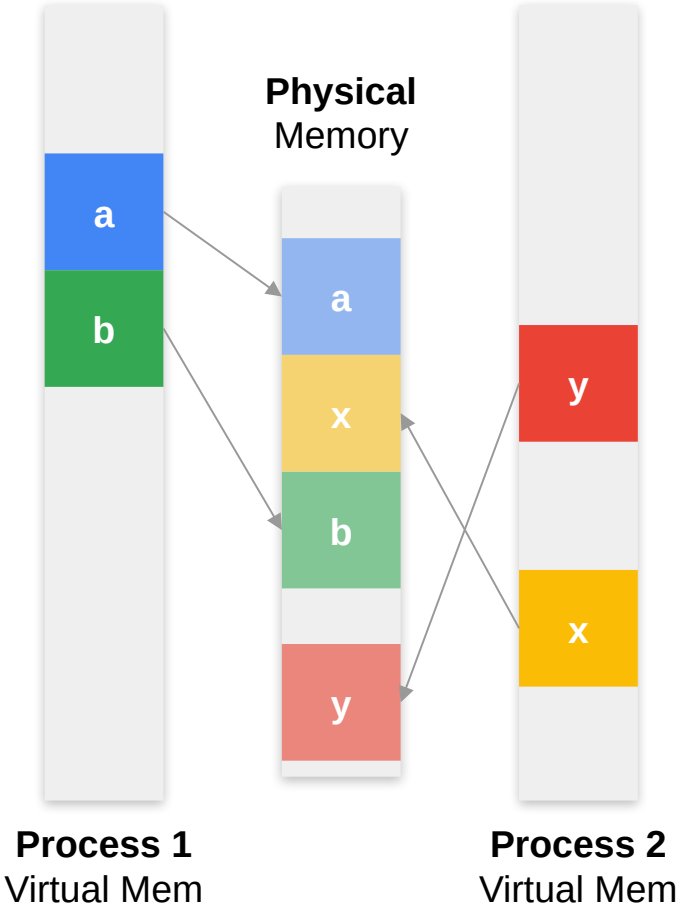
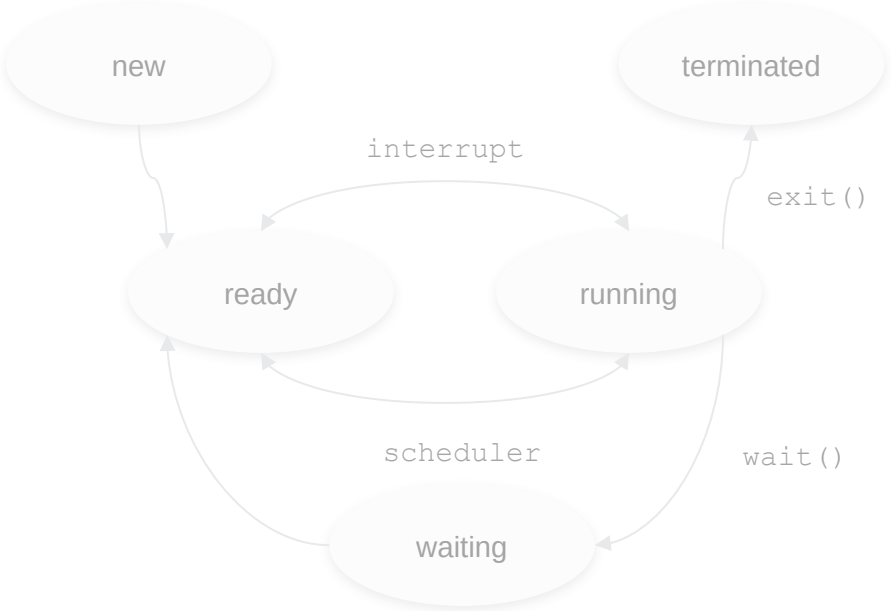
Embedded ML Software


PyTorch
TensorFlow
Caffe
scikit
learn
theano
torch

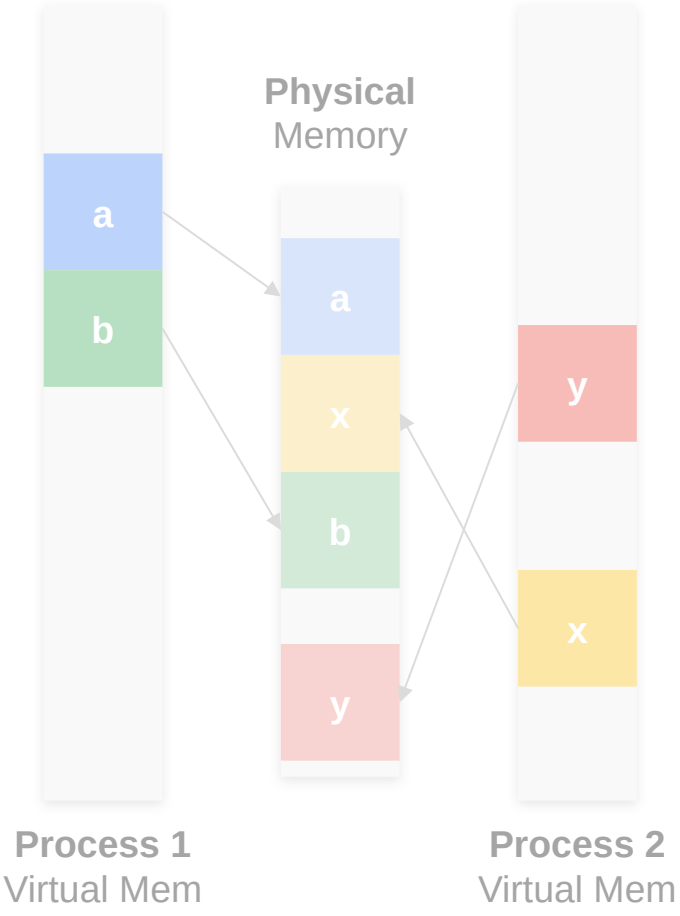

Canned Estimators
Estimators
Keras Model
Layers
Datasets
TensorFlow
Python Frontend
C++
Java
Go
...
TensorFlow Distributed Execution Engine
CPU
GPU
OS
XLA (Accelerated LinAlg Compiler)
CPU
GPU
TPU
...

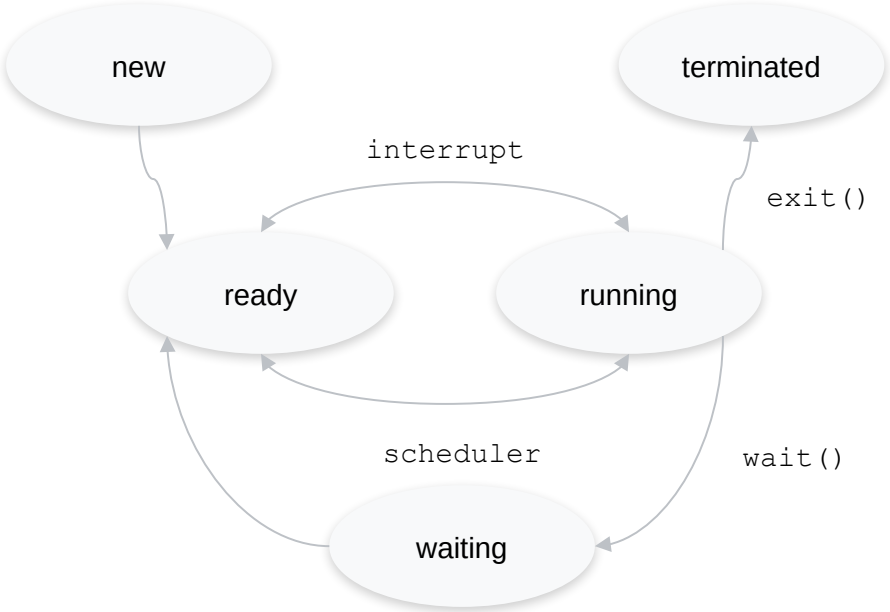

mac
OS
iOS
Linux
Mobile OS





process control





process control

Embedded Systems

Arduino
BLE Sense 33

Himax
WE-I Plus EVB

SparkFun
Edge 2

Espressif
EYE

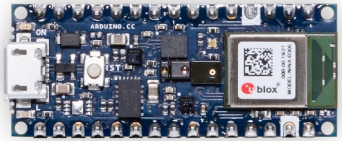
Embedded Systems

Arduino
BLE Sense 33

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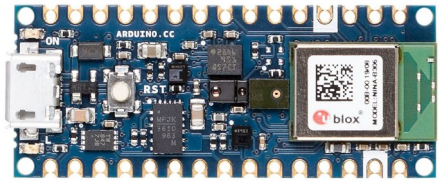
SparkFun
Edge 2

Espressif
EYE



	Micro**processor**	>	Micro**controller**
Platform			
Compute	1GHz–4GHz	~10X	1MHz–400MHz
Memory	512MB–64GB	~10000X	2KB–512KB
Storage	64GB–4TB	~100000X	32KB–2MB
Power	30W–300W	~1000X	150µW–23.5mW

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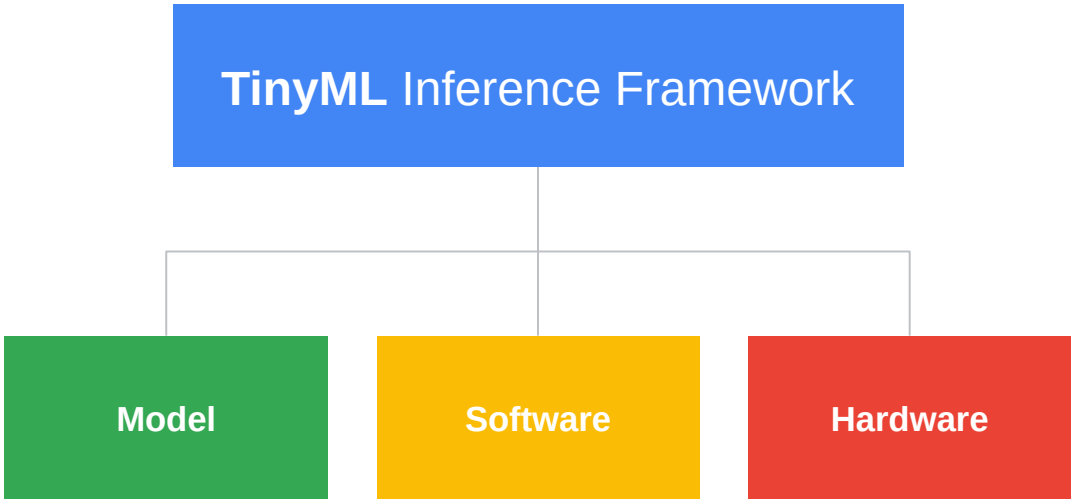
Less memory

Limited OS support

Lower compute power

Only focused on ***inference***

?


TinyML Inference Framework
Model
Software
Hardware

ML Framework
Design Checklist

Model

Support for training?	Yes	**No**
Support for inference?	**Yes**	No
How many ops?	**few**	

ML Framework
Design Checklist

Model

Support for training?	Yes	**No**
Support for inference?	**Yes**	No
How many ops?	**few**	

Software

Need quantization/optimization tools?	**Yes**	No
Can rely on virtual memory support?	Yes	**No**

ML Framework
Design Checklist

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Software

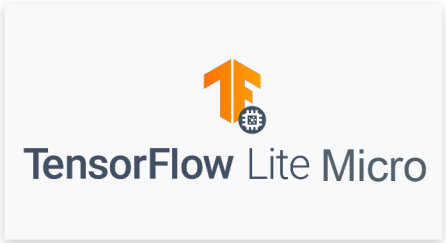
Need quantization/optimization tools?	**Yes**	No
Can rely on virtual memory support?	Yes	**No**

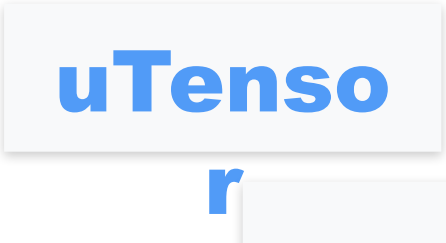
Hardware

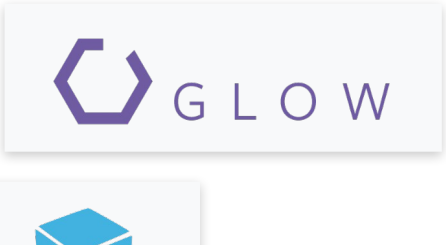
Support a diverse range of processor hardware on a device?	Yes	**No**
Need to support many different platforms and architectures?	**Yes**	No

“Tiny” Machine Learning **Frameworks**











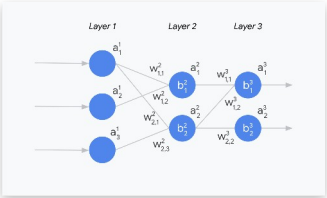
Arduino
BLE Sense 33

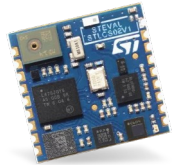
Himax
WE-I Plus EVB

SparkFun
Edge 2

Espressif
EYE

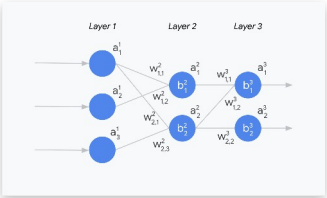
pre-trained NN model
topology, weights, bias





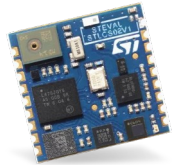
STM32L476
Cortex-M4 **MCU**

pre-trained NN model
topology, weights, bias



STM32 Cube.AI

command line interface

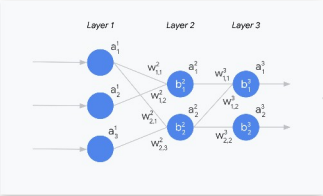


STM32L476
Cortex-M4 **MCU**

parameters:

(1) name, (2) compression factor, (3) targeted STM32

pre-trained NN model
topology, weights, bias

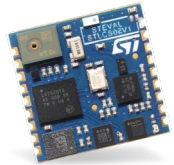


upload

Generate

STM32 Cube.AI

command line interface

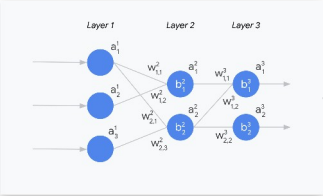


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upload

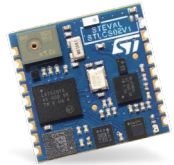
Importer

PINNR

Generate

STM32 Cube.AI

command line interface

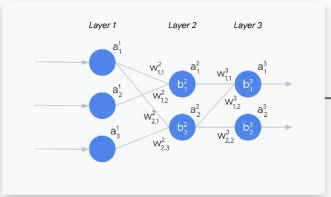


STM32L476
Cortex-M4 **MCU**

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pre-trained NN model
topology, weights, bias



upload

Importer

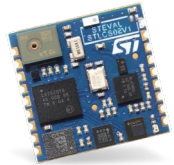
PINNR

C-code generator
(optimizer)

Generate

STM32 Cube.AI

command line interface

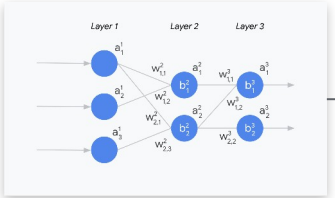


STM32L476
Cortex-M4 **MCU**

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pre-trained NN model
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upload

Importer

PINNR

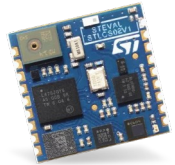
C-code generator
(optimizer)

Generate

STM32 Cube.AI

command line interface

target-dependent
optimized kernel runtime libraries

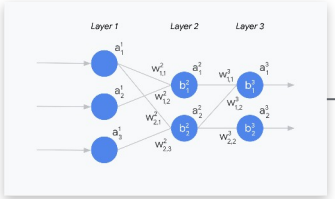


STM32L476
Cortex-M4 **MCU**

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pre-trained NN model
topology, weights, bias



upload

Importer

PINNR

C-code generator
(optimizer)

Generate

.c
.h

.a

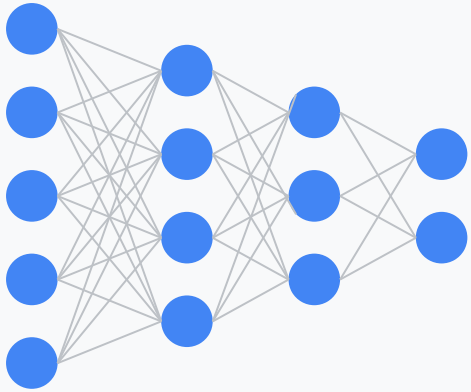
STM32 Cube.AI
command line interface

target-dependent
optimized kernel runtime libraries



STM32L476
Cortex-M4 **MCU**

Model



*3 dense **layers***
*321 **parameters***
***No** quantization or compression*

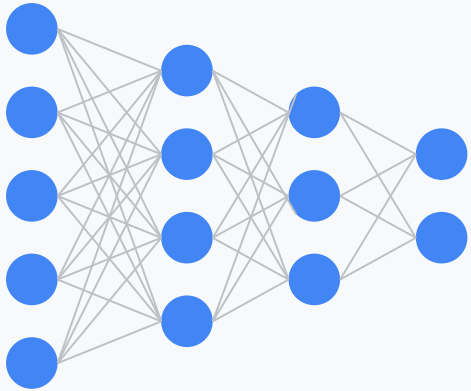
STM32 Cube.AI

27kB Flash
5kB RAM
77uS Inference Time
Closed Source

TensorFlow Lite

50kB Flash
4.7kB RAM
104uS Inference Time
Open Source

Model



*3 dense **layers***
*321 **parameters***
***No** quantization or compression*

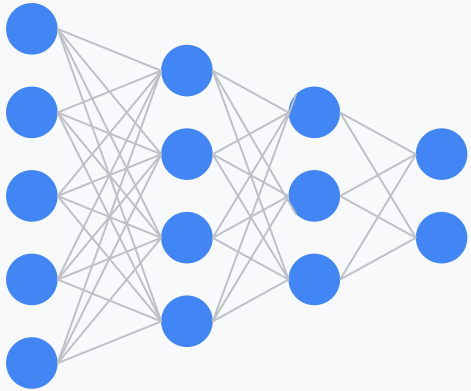
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Model



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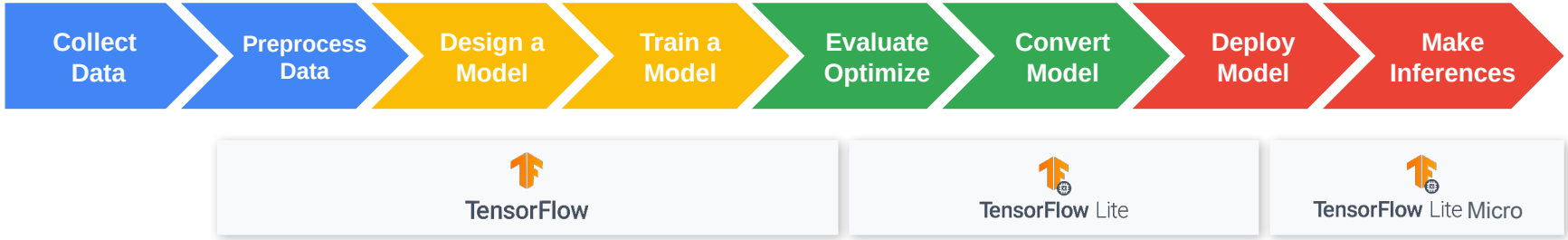
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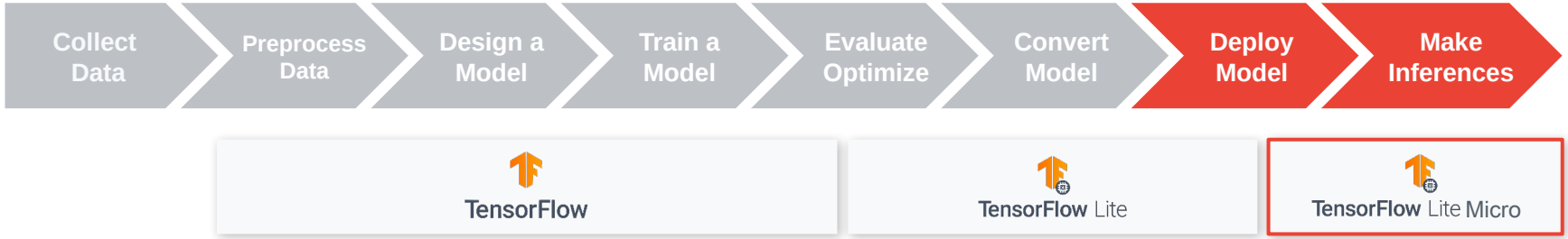
Choosing Frameworks

Hardware, IDE	Training/Embedded	other
compute **memory** constraints	**training framework** **embedded framework** (you'll export to)	**documentation** **sample code** (for use case) **personal experience**

Choosing Frameworks



Choosing Frameworks





Pete Warden, Technical Lead, TensorFlow Mobile and Embedded Team, Google.